

Algebraic Independence Forces Simultaneous Nonvanishing

A full answer to the second final-comments conjecture of arXiv:1502.01049

Candidate solution packet — run `fa_banach_001`, lane 0

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Status. Candidate full affirmative answer, likely valid, subject to human review. The result answers only the second conjecture in the source paper’s final paragraph.

1 Source and exact question

Philippe Ryckelynck and Laurent Smoch study matricial polynomial functions and a convolution-type functional equation in [1]. In their final comments (source PDF page 15), they conjecture a transcendence-degree variant of their Theorem 4.1. The source passage is reproduced below.

robust in the sense that the solution is given explicitly. Indeed, the central parts of the proofs of the four previous theorems highlight not only the well-posedness of (1.1) but also an effective computational way to construct interval by interval the solution $\mathbf{u}$.

In scope of future work, we suggest the three following problems. First, the formulas (6.4) and (6.5) imply that, in the very special case when $\mathbf{f}$ is constant, $\mathbf{u}(t)$ may be expressed through exponential-monomial functions inside $[t_0 + N\varepsilon, t_f - N\varepsilon]$. So, in this case, the search for pseudo-periodic solutions of (1.1) is linked to the difficult elimination problem of requiring that the spectrum $Sp(\mathbf{A})$ of $\mathbf{A}$ is included in the unit circle of $\mathbb{C}$. Second, we conjecture the following variant of Theorem 4.1. If the coefficients $\kappa_{m_1, m_2, \dots}$ occurring in formula (4.1) are rational or even algebraic over $\mathbb{Q}$ and, in contrast, the entries $\zeta_{i,j}$ and $\xi_{i,j}$ generate a subfield of $\mathbb{C}$ of transcendence degree equal to $2d^2$, then all the quantities $\psi_{i,j}^{(k)}(\zeta, \xi)$ and $\det(\psi^{(k)})(\zeta, \xi)$ do not vanish. Lastly, we conjecture also that for all $N \geq 2$, the genericity assumption on the multiplets in Theorems 6.1 and 6.2 hold on open dense subsets of appropriate products of matricial algebras. This conjecture seems natural in light of the two theorems presented in Section 5.

REFERENCES

- [1] Agarwal R., Bohner M., O'Regan D., Peterson A., *Dynamic equations on time scales: a survey*, J. Comput. Appl. Math. **141** (2002) 1–26 .
- [2] Hilger S., *Differential and Difference Calculus - Unified!* Nonlinear Anal. **30**(5) (1997) 2683–

Figure 1: Full-width crop of source PDF page 15. The target is the second conjecture, beginning “Second, we conjecture.”

The conjecture is read with the hypotheses inherited from the phrase “variant of Theorem 4.1”: one is given finitely many *generic* matricial polynomial functions

$$\psi^{(k)}(\zeta, \xi) = \sum_m \kappa_m^{(k)} \zeta^{m_1} \xi^{m_2} \zeta^{m_3} \xi^{m_4} \dots, \quad 1 \leq k \leq \ell.$$

If all coefficients are rational, or more generally algebraic over $\mathbb{Q}$, and the $2d^2$ entries of $\zeta, \xi \in M_d(\mathbb{C})$ generate a subfield of $\mathbb{C}$ of transcendence degree $2d^2$, must every $\psi_{ij}^{(k)}(\zeta, \xi)$ and every $\det \psi^{(k)}(\zeta, \xi)$ be nonzero?

The inherited genericity is essential: with no nonzero/generic hypothesis, the zero polynomial is an immediate counterexample. The source's definition of genericity is exactly the one used in Theorem 4.1 and its proof [1, pp. 6–7].

2 Definitions and proof intuition

After expanding matrix products, each scalar entry of $\psi^{(k)}(\zeta, \xi)$ is an ordinary commutative polynomial in the $2d^2$ scalar entries of (ζ, ξ) ; so is its determinant. The proof of source Theorem 4.1 shows that genericity makes every one of these formal polynomials nonzero.

The remaining step is field-theoretic. A tuple of n elements generating a field of transcendence degree n is algebraically independent. Moreover, algebraically independent elements remain so after an algebraic extension of the base field. Here $n = 2d^2$, and the finitely many algebraic coefficients generate just such an algebraic extension of $\mathbb{Q}$. Thus none of the source paper's nonzero polynomials can vanish at the prescribed matrix pair.

3 Formal result

Lemma 1 (Algebraic base change). *Let K/F be an algebraic field extension and let $x = (x_1, \dots, x_n)$ lie in a common overfield. If $\text{trdeg}_F F(x) = n$, then $x_1, \dots, x_n$ are algebraically independent over K .*

Proof. Since K/F is algebraic, adjoining K does not change transcendence degree:

$$\text{trdeg}_K K(x) = \text{trdeg}_F F(x) = n.$$

If the n elements $x_1, \dots, x_n$ were algebraically dependent over K , a transcendence basis could be chosen from fewer than n of them, forcing $\text{trdeg}_K K(x) < n$, a contradiction. $\square$

Theorem 1 (Affirmative answer to the second conjecture). *Let $\psi^{(1)}, \dots, \psi^{(\ell)}$ be generic matricial polynomial functions in the sense of Ryckelynck–Smoch Theorem 4.1. Suppose every coefficient $\kappa_m^{(k)}$ is algebraic over $\mathbb{Q}$. Let $\zeta, \xi \in M_d(\mathbb{C})$, and suppose the field generated over $\mathbb{Q}$ by their $2d^2$ scalar entries has transcendence degree $2d^2$. Then, for all $1 \leq k \leq \ell$ and $1 \leq i, j \leq d$,*

$$\psi_{ij}^{(k)}(\zeta, \xi) \neq 0, \quad \det(\psi^{(k)}(\zeta, \xi)) \neq 0.$$

In particular, the conjecture holds for rational coefficients.

Proof. List the entries of (ζ, ξ) as $x = (x_1, \dots, x_n)$, where $n = 2d^2$, and set $E = \mathbb{Q}(x)$. The hypothesis $\text{trdeg}_{\mathbb{Q}} E = n$ implies that the n -tuple x is algebraically independent over $\mathbb{Q}$: any algebraic dependence would reduce the maximal size of an algebraically independent subtuple below n .

Let K be the field generated over $\mathbb{Q}$ by all coefficients occurring in the finite family $\psi^{(1)}, \dots, \psi^{(\ell)}$. There are only finitely many such coefficients, and each is algebraic over $\mathbb{Q}$, so $K/\mathbb{Q}$ is a finite algebraic extension. By the lemma, $x_1, \dots, x_n$ are algebraically independent over K .

For every i, j, k , expansion of the matrix words gives an ordinary polynomial

$$P_{ij}^{(k)}(X_1, \dots, X_n) \in K[X_1, \dots, X_n]$$

whose value at x is $\psi_{ij}^{(k)}(\zeta, \xi)$. Likewise,

$$D^{(k)}(X_1, \dots, X_n) := \det(P_{ij}^{(k)}(X_1, \dots, X_n))_{i,j=1}^d \in K[X_1, \dots, X_n].$$

The proof of source Theorem 4.1 establishes, from the source’s genericity conditions, that every $P_{ij}^{(k)}$ and every $D^{(k)}$ is a nonzero formal polynomial [1, pp. 6–7]. (For determinants, the source specializes $\zeta = \rho I_d$, $\xi = \sigma I_d$, obtaining the d -th power of a nonzero scalar polynomial; its upper-triangular specialization and conjugation argument handles all entries.)

Algebraic independence of x over K means precisely that the evaluation homomorphism

$$K[X_1, \dots, X_n] \longrightarrow K(x_1, \dots, x_n), \quad P \longmapsto P(x),$$

is injective. Therefore no $P_{ij}^{(k)}$ and no $D^{(k)}$ can evaluate to zero. These evaluations are exactly the quantities asserted to be nonzero, completing the proof. $\square$

4 Verification, scope, and novelty

The proof uses no analytic regularity, recurrence theory, or computational assumption. It requires only (i) the nonzero-formal-polynomial conclusion proved inside source Theorem 4.1 and (ii) invariance of transcendence degree under algebraic base extension. The conclusion is simultaneous for every member of the source’s finite family.

The result does *not* address the source’s first pseudo-periodicity problem or its third conjecture on open-dense genericity for arbitrary $N \geq 2$. It also does not support a reading that drops genericity; that reading is false for $\psi = 0$.

A bounded search on 27 August 2026 covered the run indexes, the exact arXiv id, title and authors, the exact transcendence-degree phrase, and close keyword variants. It found the source and journal/HAL copies but no separate paper answering this conjecture. Novelty confidence is moderate rather than exhaustive; no full MathSciNet/zblMATH citation-graph search was available.

Human-review focus. Verify that the phrase “variant of Theorem 4.1” retains the genericity hypothesis, and check the algebraic base-change lemma. Under that natural and necessary reading, the argument is complete.

References

- [1] P. Ryckelynck and L. Smoch, *On the functional equation $\alpha \mathbf{u} + \mathcal{C} \star (\chi \mathbf{u}) = \mathbf{f}$* , North-Western European Journal of Mathematics **4** (2018), 1–19; arXiv:1502.01049.